**2** **(a)** Taking moments about X at the wall, and letting the length be L,
Sum of clockwise moments = sum of anti-clockwise moments 1

$$W\times\left(\frac{1}{2}L\cos 10^\circ\right)=T\times\left(\frac{1}{4}L\sin 60^\circ\right)$$

$$3.0\times\left(\frac{1}{2}\cos 10^\circ\right)=T\times\left(\frac{1}{4}\sin 60^\circ\right)$$ 1

$T$ = 6.823 N = 6.8 N

**(b)** **(i)**

$$k=\frac{T}{e}$$

$$k=\frac{6.8}{(0.38-0.30)}$$ 1

$$=85\text{ N m}^{-1}$$ 1

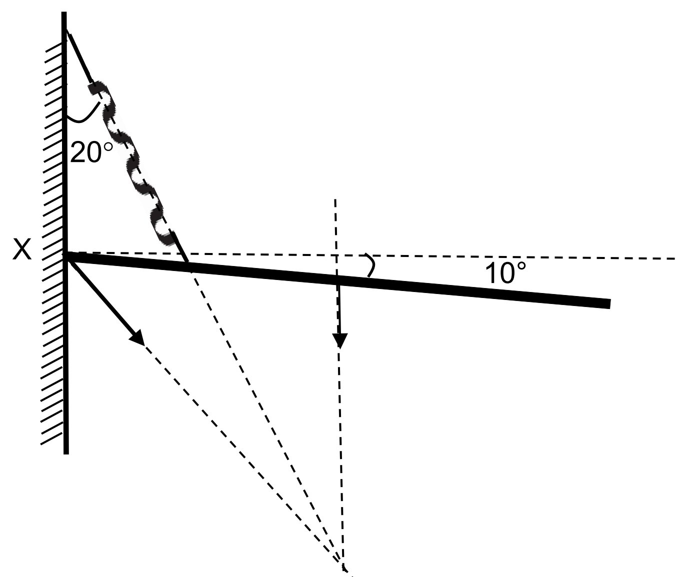
**(ii)** **1.**

$$\frac{\Delta k}{k}=\frac{\Delta T}{T}+\frac{\Delta e}{e}=\frac{\Delta T}{T}+\frac{\Delta(L-L_0)}{(L-L_0)}$$

$$\frac{\Delta k}{k}=0.02+\frac{0.2}{8.0}$$ 1

$$\frac{\Delta k}{k}=0.045=4.5\%$$ 1

**(c)** **(i)** 1




**(ii)** Let the force be $F$.

$$F_X=6.8\sin 20^\circ=2.326\text{ N}$$

$$F_Y=6.8\cos 20^\circ-3.0=3.390\text{ N}$$ 1

$$F=\sqrt{2.326^2+3.390^2}=4.1\text{ N}$$ 1

[Total: 9]

3 (a) From point of release to point of maximum compression